# B. Minimize Equal Sum Subarrays

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: constructive algorithms, math, number theory |
https://codeforces.com/problemset/problem/1998/B

*It is known that Farmer John likes Permutations, but I like them too!*

— Sun Tzu, *The Art of Constructing Permutations*

You are given a permutation* $p$ of length $n$.

Find a permutation $q$ of length $n$ that minimizes the number of pairs $(i, j)$ $(1 \le i \le j \le n)$ such that $p_i + p_{i+1} + \ldots + p_j = q_i + q_{i+1} + \ldots + q_j$.

*A permutation of length $n$ is an array consisting of $n$ distinct integers from 1 to $n$ in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation ($n$=3 but there is 4 in the array).

## Input

The first line contains $t$ $(1 \le t \le 10^4)$ — the number of test cases.

The first line of each test case contains $n$ $(1 \le n \le 2 \cdot 10^5)$.

The following line contains $n$ space-separated integers $p_1, p_2, \ldots, p_n$ $(1 \le p_i \le n)$ — denoting the permutation $p$ of length $n$.

It is guaranteed that the sum of $n$ over all test cases does not exceed $2 \cdot 10^5$.

## Output

For each test case, output one line containing any permutation of length $n$ (the permutation $q$) such that $q$ minimizes the number of pairs.

## Example

| Input |
| --- |
| 3<br>2<br>1 2<br>5<br>1 2 3 4 5<br>7<br>4 7 5 1 2 6 3 |

| Output |
| --- |
| 2 1<br>3 5 4 2 1<br>6 2 1 4 7 3 5 |

## Note

For the first test, there exists only one pair $(i, j)$ $(1 \leq i \leq j \leq n)$ such that $p_i + p_{i+1} + \ldots + p_j = q_i + q_{i+1} + \ldots + q_j$, which is (1, 2). It can be proven that no such $q$ exists for which there are no pairs.